Electronic Companion: The Shared Discovery Paradox

This companion uses the main manuscript's notation and its nondegenerate domain $M \geq 2$, $2 \leq N \leq M$, $1/M < p < 1$, unless a boundary case is stated explicitly. Section numbers EC.1–EC.6 refer to this document; other labels refer to the main manuscript.

EC.1 Exact independent enumeration

By symmetry, condition on box 1 being the target. The report-count vector

$$C = (C_1, \ldots, C_M) \sim \text{Multinomial}(N; p, q, \ldots, q).$$

For a count vector x, let

$$r(x) = |\{b : x_b > x_1\}|, \qquad t(x) = |\{b : x_b = x_1\}|.$$

With uniform randomization at cutoff ties, the conditional probability that the true box belongs to a top-L count set is

$$\tau_L(x) = \begin{cases} 0, & r(x) \geq L, \\ 1, & r(x) + t(x) \leq L, \\ \dfrac{L - r(x)}{t(x)}, & r(x) < L < r(x) + t(x). \end{cases} \tag{EC.1}$$

Hence the exact pooled frontier is

$$G_{M,N,L}(p) = \sum_{x_1 + \cdots + x_M = N} \frac{N!}{\prod_{b=1}^{M} x_b!} p^{x_1} q^{N - x_1} \tau_L(x). \tag{EC.2}$$

For $M = 16$ and $N = 8$, the sum contains

$$\binom{N+M-1}{M-1} = \binom{23}{15} = 490{,}314$$

count vectors. Grouping by permutations of the fifteen non-target boxes reduces both the independent and common-cue calculations to 67 count classes.

At $L = 7$, the difference between the pooled portfolio and private clue-following can be decomposed as

$$\mathbb{P}(\theta \in S_7^* \setminus S_{\text{private}}) - \mathbb{P}(\theta \in S_{\text{private}} \setminus S_7^*).$$

Exact enumeration gives a synthesis gain of 0.011744 and a compression loss of 0.007962. At $L = 8$, no box named by a private clue is discarded; the improvement 0.027193 comes entirely from reallocating duplicated capacity.

EC.2 Common-cue likelihood and Bayes ranking

For a candidate target t, define the report probability vector

$$\rho_r^{(t)} = \begin{cases} p, & r = t, \\ q, & r \neq t. \end{cases}$$

Let z be the common cue and k the copier count. For observed counts x,

$$\mathbb{P}(C = x \mid \theta = t, X_0 = z, K = k) \tag{EC.3}$$

$$= \mathbf{1}\{x_z \geq k\} \frac{(N-k)!}{(x_z-k)! \prod_{r\neq z} x_r!} \prod_{r=1}^{M} \left(\rho_r^{(t)}\right)^{x_r - k\mathbf{1}\{r=z\}}. \tag{EC.4}$$

Marginalizing gives

$$\mathbb{P}(C = x \mid \theta = t, c) = \sum_{z=1}^{M} \rho_z^{(t)} \sum_{k=0}^{N} \binom{N}{k} c^k (1-c)^{N-k} \mathbb{P}(C = x \mid \theta = t, X_0 = z, K = k). \tag{EC.5}$$

With a uniform prior, normalize these M likelihoods to obtain the herd-aware posterior. The verification script precomputes the coefficient of every basis term $c^k(1-c)^{N-k}$ for each count class and candidate target.

EC.3 Proof of the monotone planner-gain theorem

We give the omitted details for equation (24) and monotonicity.

Lemma EC.1 (Named states precede unnamed states)**.** *For every realizable report vector under the common-cue model, every state with a positive report count has posterior probability at least as large as every state with zero reports.*

Proof. Fix a named state b and an unnamed state d. Compare the mixture likelihood under hypotheses $\theta = b$ and $\theta = d$ term by term in equation (EC.5). If the common cue is neither b nor d, the cue prefactor is identical while the residual multinomial likelihood differs by a nonnegative power of p/q in favor of b. If the cue is b, its prefactor is p under $\theta = b$ and q under $\theta = d$, and the residual likelihood contributes another nonnegative power of p/q. If the cue is d, feasibility requires $k = 0$ because d is unnamed; the combined likelihood ratio is $(p/q)^{x_b - 1} \geq 1$. Every matched nonzero term weakly favors the named state. Summing over cues and copier counts preserves the inequality. □

The optimal N-action portfolio therefore includes all D distinct named states and fills $N-D$ positions with unnamed states. On the event that private report-following misses, the target lies among the $M - D$ unnamed states. Symmetry assigns equal conditional posterior mass to those states, so the planner's conditional rescue probability is $(N - D)/(M - D)$. When $D = M$, the miss event is impossible; the product in equation (24) is defined as zero rather than evaluating $0/0$. This proves equation (24).

Condition on copier count $K = k$. Given k, the joint law of the common cue and the $N - k$ independent clues does not otherwise depend on c. Define

$$h(k) = \mathbb{E}\left[\mathbf{1}\{\text{miss}\}\frac{N - D}{M - D}\middle| K = k\right].$$

Then $\Delta_{\text{planner}}(c) = \mathbb{E}[h(K)]$ for $K \sim \text{Binomial}(N, c)$. We have $h(0) = h(1)$ because one copy of X_0 has the same joint distribution as one independent clue. For $k \geq 1$, couple the k-copier system and the $(k+1)$-copier system by converting one independent searcher into an additional copier while holding the common cue and all other reports fixed. Because the common cue is already represented, the conversion can only weakly reduce D. It can only weakly increase the miss indicator. The fraction $(N - D)/(M - D)$ is nondecreasing as D falls when $N \leq M$. Thus $h(k + 1) \geq h(k)$. For $1/M < p < 1$ and $N \geq 2$, the inequality is strict at $k = N - 1$: with positive probability, the converted independent clue names the target (probability $p > 0$) while the already represented common cue misses (probability $1 - p > 0$). Before conversion, the miss

integrand is zero; after conversion, $D = 1$ and it equals $(N-1)/(M-1) > 0$. This event has positive probability $p(1-p)$, giving $h(N) > h(N-1)$.

If $K' \sim \text{Binomial}(N-1, c)$, the standard binomial derivative identity yields

$$\frac{d}{dc}\mathbb{E}[h(K)] = N\mathbb{E}[h(K'+1) - h(K')] > 0 \qquad (c \in (0,1)).$$

At $p = 1$, both protocols succeed surely and the gain is zero, so strictness requires the stated upper bound on p. The hypothesis $N \geq 2$ is also necessary: with a single searcher the named box is the posterior mode, the planner replicates private search, the gain is identically zero, and the derivative vanishes throughout $(0,1)$ because $h(0) = h(1)$. At $c = 1$, one box is named. The planner adds $N-1$ of the $M-1$ unnamed boxes, each with posterior mass q, while private search covers only the named box. Hence $\Delta_{\text{planner}}(1) = (N-1)q$.

EC.4 Proof of the proportional sparse-evidence limit

Throughout, $N = N_M$ with $N_M/M \to \alpha \in (0,1)$, and $p_M = r/(M-1+r)$, $q_M = 1/(M-1+r)$ for fixed $r > 1$, so that $N_M p_M \to \alpha r$ and $N_M q_M \to \alpha$: the target receives an asymptotically Poisson(αr) number of reports, while each false box receives approximately Poisson(α). All limits are as $M \to \infty$ and all clues are independent.

Blind portfolio. The coordinator selects N_M distinct boxes without observing clues. Under the uniform prior, $G_{\text{blind}} = N_M/M \to \alpha$.

Private clue-following. By equation (1), $G_{\text{private}} = 1-(1-p_M)^{N_M}$, and $N_M \log(1-p_M) = -N_M p_M\big(1 + O(p_M)\big) \to -\alpha r$, so $G_{\text{private}} \to 1 - e^{-\alpha r}$.

Consensus. Condition on θ. The target count $C_\theta \sim \text{Binomial}(N_M, p_M)$ converges in distribution to Poisson(αr) and is therefore bounded in probability. Conditional on the number of off-target reports $F = N_M - C_\theta$, the false counts are multinomial with F trials over $M-1$ equally likely boxes, and $F/M \to \alpha$ in probability. Fix an integer j and let $W_j = |\{b \neq \theta : C_b \geq j\}|$. Standard occupancy asymptotics give $\mathbb{E}[W_j]/M \to \mathbb{P}(\text{Poisson}(\alpha) \geq j) > 0$, and the variance of W_j is $O(M)$ because the occupancy indicators are negatively correlated, so $W_j \to \infty$ in probability. Hence $\max_{b\neq\theta} C_b \geq j$ with probability tending to one for every fixed j: the largest false count diverges in probability while C_θ remains tight. Consensus success requires $C_\theta \geq \max_{b\neq\theta} C_b$ (an event whose probability tends to zero), so $G_{\text{consensus}} \to 0$; uniform tie-breaking only lowers the success probability further.

Portfolio. By equation (8), every named box has strictly larger posterior mass than every unnamed box,

and at most N_M boxes are named, so the top-N_M rule retains every named box and fills its remaining slots uniformly at random among the posterior-tied unnamed boxes. Let D be the number of distinct named boxes. On $\{C_\theta \geq 1\}$ the portfolio contains θ. On the miss event $\{C_\theta = 0\}$, the target is one of the $M - D$ unnamed boxes, which carry equal conditional posterior mass, so the conditional rescue probability is $(N_M - D)/(M - D)$. This is the $c = 0$ case of equation (24):

$$G_{\text{portfolio}} = G_{\text{private}} + \mathbb{E}\left[\mathbf{1}\{C_\theta = 0\}\,\frac{N_M - D}{M - D}\right].$$

Conditional on the miss event, D is the number of occupied boxes when N_M balls are placed uniformly over $M - 1$ boxes. Occupancy concentration ($\mathbb{E}[D] = (M-1)[1 - (1 - \frac{1}{M-1})^{N_M}]$ with variance $O(M)$) gives $D/M \to 1 - e^{-\alpha}$ in probability, hence

$$\frac{N_M - D}{M - D} \;\longrightarrow\; \tau_\alpha = \frac{\alpha - (1 - e^{-\alpha})}{e^{-\alpha}} = 1 - (1 - \alpha)e^{\alpha} \qquad \text{in probability,}$$

where $\tau_\alpha \in (0, 1)$ because $e^{-\alpha} > 1 - \alpha$ for $\alpha > 0$: duplicated reports leave free slots asymptotically. The ratio is bounded by one, so bounded convergence together with $\mathbb{P}(C_\theta = 0) \to e^{-\alpha r}$ yields

$$G_{\text{portfolio}} \longrightarrow \left(1 - e^{-\alpha r}\right) + e^{-\alpha r}\left[1 - (1 - \alpha)e^{\alpha}\right] = 1 - (1 - \alpha)\, e^{-\alpha(r-1)}.$$

Finally, $(1 - \alpha)e^{\alpha}$ is strictly decreasing on $(0, 1)$ from the value one at $\alpha = 0$, so $(1 - \alpha)e^{\alpha} < 1$ and the portfolio limit strictly exceeds the private limit for every $\alpha \in (0, 1)$. □

Symmetric equal-split market. Write the scaled posterior weight of box b as $W_b = M\pi_b = r^{C_b}/(Z_M/M)$ with $Z_M = \sum_b r^{C_b}$, and recall from proposition EC.1 the notation $f_j = e^{-\alpha}\alpha^j/j!$, $\psi_j = r^j e^{-\alpha(r-1)}$, $g_j = f_j\psi_j = e^{-\alpha r}(\alpha r)^j/j!$, and $h(x) = (1 - e^{-x})/x$.

Step 1: weighted occupancy law of large numbers. For a false box, $\mathbb{E}\left[r^{C_b}\right] = \left(1 + (r-1)q_M\right)^{N_M} \to e^{\alpha(r-1)}$ and $\mathbb{E}\left[r^{2C_b}\right] = \left(1 + (r^2-1)q_M\right)^{N_M} \to e^{\alpha(r^2-1)} < \infty$. For two false boxes $a \neq b$, each clue lands on at most one of them, so $\mathbb{E}\left[r^{C_a} r^{C_b}\right] = \left(1 + 2(r-1)q_M\right)^{N_M}$, and a direct expansion shows the covariance is $O(1/M)$ times a bounded factor. Chebyshev's inequality then gives $Z_M/M \to e^{\alpha(r-1)}$ in probability, so $W_b \to \psi_j$ in probability for a box with $C_b = j$; the target term has bounded expectation $\mathbb{E}[r^{C_\theta}] = (1 + (r-1)p_M)^{N_M} = O(1)$ and hence contributes $o_{\mathbb{P}}(1)$ after division by M. For false-box tails, $r^{C_b}\mathbf{1}\{C_b \geq J\} \leq r^{-J} r^{2C_b}$, so their normalized tail expectation is uniformly $O(r^{-J})$; Markov's inequality makes the tails uniformly small in probability. Finite count-class frequencies converge by occupancy concentration; truncation then yields, for the weighted sums, $M^{-1}\sum_b r^{C_b}\mathbf{1}\{C_b \geq J\} \to e^{\alpha(r-1)}\sum_{j \geq J} g_j$ in probability,

with the tail uniformly small in J because $\sum_j g_j = 1$ (the identity $f_j\psi_j = g_j$ is the exponential tilting of a Poisson law).

Step 2: convergence of the budget equation. The same tail bound gives $\max_b W_b/M \to 0$ in probability; thus for any fixed $\Lambda > 0$ the lower inverse endpoint $\max_b W_b/N_M$ tends to zero. The finite inverse/budget equation is therefore in its non-pure domain with probability tending to one. At the finite anonymous equilibrium of proposition 5, an active box satisfies $W_b\,\phi(s_b) = \Lambda_M := M\lambda_M$, and the budget constraint $\sum_b s_b = 1$ reads $M^{-1}\sum_b x_b = N_M/M$ with $x_b = N_M s_b$. Since $(1 - x/N)^N \to e^{-x}$ uniformly on compact x-sets, $\phi(x/N) \to h(x)$ uniformly on compacts, with h continuous and strictly decreasing from $h(0) = 1$. Define the limit budget function

$$B(\Lambda) = \sum_{j\,:\,\psi_j > \Lambda} f_j\, h^{-1}\big(\Lambda/\psi_j\big).$$

Because $h(x) \le 1/x$, every active class obeys $x_j \le \psi_j/\Lambda$, so $B(\Lambda) \le \Lambda^{-1}\sum_j g_j = 1/\Lambda$ and the series converges locally uniformly; B is therefore continuous, strictly decreasing where positive, with $B(0^+) = \infty$ and $B(\infty) = 0$, and equation (EC.6) has a unique root Λ. The same bound $x_b \le W_b/\Lambda$ controls the high-count contribution to the finite budget uniformly in M by Step 1, so the finite (random) budget functions converge to B in probability at every fixed Λ. Strict monotonicity then squeezes the equilibrium value: for any $\Lambda' < \Lambda < \Lambda''$, eventually $B_M(\Lambda') > N_M/M > B_M(\Lambda'')$ with probability tending to one, so $\Lambda_M \to \Lambda$ in probability.

Step 3: from values to discovery. The welfare identity of proposition 6 gives conditional discovery $N_M\lambda_M = (N_M/M)\Lambda_M \le 1$, so $\Lambda_M \le M/N_M$ is uniformly bounded, and bounded convergence yields $G^{\mathrm{sym}}_{\mathrm{market}} = \mathbb{E}[N_M\lambda_M] \to \alpha\Lambda$.

Step 4: the tilted form. Multiplying the budget equation by Λ and using $\Lambda = \psi_j h(x_j)$ on active classes,

$$\alpha\Lambda = \sum_j f_j x_j\,\psi_j h(x_j) = \sum_j f_j\psi_j\big(1 - e^{-x_j}\big) = \sum_j g_j\big(1 - e^{-x_j}\big)\,:$$

in the limit the market covers the target's own count class j with probability $1 - e^{-x_j}$, where the target's count is Poisson(αr) and the number of occupants of any fixed box is asymptotically Poisson(x_j). Canonically $\Lambda = 1.094021$ lies strictly between $\psi_1 = 0.948$ and $\psi_2 = 3.555$, so the limit market abandons every box with fewer than two reports. □

Abundant evidence (remark EC.1). Hold p fixed. By the multiplicative Chernoff bound, $\mathbb{P}(C_\theta \le N_M p/2) \le e^{-N_M p/8} \to 0$. Each false count is Binomial(N_M, q_M) with mean $N_M q_M \to \alpha(1 - p)$, so the standard upper-tail bound gives $\mathbb{P}(C_b \ge N_M p/2) \le \big(2eq_M/p\big)^{N_M p/2}$, whose base is $O(1/M)$ while

the exponent grows linearly in M; the union bound over the $M-1$ false boxes therefore vanishes. With probability tending to one the target is the unique plurality winner, so $G_{\text{consensus}} \to 1$. The same argument applies whenever $\lambda_M = N_M p_M$ satisfies $\lambda_M / \log M \to \infty$: the target count concentrates above $\lambda_M/2$ while $M \exp\left[-(\lambda_M/2)\log(\lambda_M/2e\alpha)\right] \to 0$. □

EC.5 The proportional sparse-evidence limit

The sixteen-box environment is deliberately small, so it is natural to ask whether the ordering is a small-number artifact. Scale the environment proportionally: let the number of searchers grow with the number of boxes, $N_M/M \to \alpha \in (0,1)$, and hold the clue likelihood ratio $r = p_M/q_M > 1$ fixed, so that

$$p_M = \frac{r}{M-1+r}, \qquad q_M = \frac{1}{M-1+r}.$$

The canonical benchmark corresponds to $\alpha = 1/2$ and $r = 3.75$. Fixing r is one natural invariant among several—one could instead preserve mutual information—but it keeps the report-count occupancy problem nondegenerate: the target receives an asymptotically Poisson(αr) number of reports, while a typical false box receives approximately Poisson(α).

Proposition EC.1 (Proportional sparse-evidence limit)**.** *In the independent symmetric model with $N_M/M \to \alpha \in (0,1)$ and fixed likelihood ratio $r > 1$, consider the blind coordinated portfolio (N_M distinct boxes chosen without clues), pooled one-action consensus, private clue-following, and the pooled top-N_M portfolio. As $M \to \infty$,*

$$G_{\text{blind}} \longrightarrow \alpha, \qquad G_{\text{consensus}} \longrightarrow 0, \qquad G_{\text{private}} \longrightarrow 1 - e^{-\alpha r}, \qquad G_{\text{portfolio}} \longrightarrow 1 - (1-\alpha)\, e^{-\alpha(r-1)}.$$

The anonymous symmetric equal-split market on the pooled posterior also converges:

$$G^{\text{sym}}_{\text{market}} \longrightarrow \alpha\Lambda, \qquad \textit{where } \Lambda \textit{ uniquely solves} \quad \sum_{j\,:\,\psi_j > \Lambda} f_j\, h^{-1}\big(\Lambda/\psi_j\big) = \alpha, \tag{EC.6}$$

with $f_j = e^{-\alpha}\alpha^j/j!$, $\psi_j = r^j e^{-\alpha(r-1)}$, and $h(x) = (1-e^{-x})/x$ strictly decreasing from $h(0) = 1$. Writing $x_j = h^{-1}(\Lambda/\psi_j)$ on active count classes and $g_j = f_j\psi_j = e^{-\alpha r}(\alpha r)^j/j!$, the equivalent tilted form is $G^{\text{sym}}_{\text{market}} \to \sum_j g_j\big(1 - e^{-x_j}\big)$.

The proof is in section EC.4. For the canonical scaling $\alpha = 1/2$, $r = 3.75$, the solution of equation (EC.6)

is $\Lambda = 1.094021$, and the complete limit ordering is

$$G_{\text{consensus}} \to 0 \;<\; G_{\text{blind}} \to 0.50 \;<\; G^{\text{sym}}_{\text{market}} \to 0.547011,$$
$$G^{\text{sym}}_{\text{market}} \to 0.547011 \;<\; G_{\text{private}} \to 0.846645 \;<\; G_{\text{portfolio}} \to 0.873580.$$

Because $(1-\alpha)e^{\alpha} < 1$ on $(0,1)$, the portfolio limit strictly exceeds the private limit at every α. The market's position strictly above consensus and weakly below the portfolio also holds at every (α, r), since the finite inequalities pass to the limit. The positions of the blind benchmark relative to private search and to the market, by contrast, depend on (α, r), and the displayed ordering is a property of the canonical scaling.

The market limit is water-filling against a Poisson count profile: in the limit the equilibrium abandons every box with fewer than two reports (canonically $\psi_1 = 0.948 < \Lambda < \psi_2 = 3.555$) and spreads over the high-count classes in which the target, whose own count is Poisson(αr), is hidden. The finite canonical value 0.599099 sits above its limit 0.547011, and the seeded Monte Carlo diagnostic in the verification script shows a monotone decline through $M \in \{16, 64, 256, 1024\}$ (0.5991, 0.5740, 0.5582, 0.5494): as the posterior spreads over more comparably ranked boxes, symmetric mixing wastes more of its budget on collisions inside the leading count classes.

Two readings of the canonical limit deserve note. First, information without assignment and assignment without information deliver nearly the same discovery: the market reaches 0.547 holding the full pooled posterior but no coordination device, the blind portfolio reaches 0.500 holding a coordination device but no information, and only their combination reaches 0.874. Second, the market–private gap at scale admits an exact accounting. Since $G_{\text{private}} \to \mathbb{P}\big(\text{Poisson}(\alpha r) \geq 1\big)$ and the canonical limit market abandons the zero- and one-report classes,

$$G_{\text{private}} - G^{\text{sym}}_{\text{market}} \;\longrightarrow\; g_1 + \sum_{j \geq 2} g_j \, e^{-x_j}$$
$$= 0.287541 + 0.012094.$$

essentially the entire gap is the probability that the target receives exactly one report. The singleton discoveries that private clue-followers keep for free are precisely what an anonymous market rationally abandons; collisions inside the classes the market does cover cost only 0.012.

Remark EC.1 (Abundant-evidence regime). The paradox is a sparse-evidence phenomenon. Hold $p \in (0,1)$ fixed with $N_M/M \to \alpha \in (0,1)$, so that the likelihood ratio $r_M = p/q_M \to \infty$. Then $G_{\text{consensus}} \to 1$, and a fortiori every clue-using protocol converges to one: the paradox vanishes in levels, and distinctions among diversified protocols persist only in convergence rates. The clean dial is $\lambda_M = N_M p_M$, the target's expected

report count: proposition EC.1 is the bounded-λ regime, while whenever $\lambda_M/\log M \to \infty$ the union bound in section EC.4 gives consistency. The intermediate window $\lambda_M \asymp \log M$ is a multiple-comparisons threshold problem we leave open. The benchmark is aimed accordingly: consensus institutions are adequate where evidence about the leading candidate is abundant; the compression loss binds in sparse-evidence search—frontier science, early-stage investing, novel-hypothesis generation—where the target's support never separates from the noise.

The limit sharpens the finite benchmark in two ways. First, the paradox is not a small-number artifact: consensus discovery vanishes while every diversified protocol retains a strictly positive limit. Second, large-scale consensus fails through an additional channel. The target's report count remains stochastically bounded, but the maximum count among linearly many false boxes diverges, so the plurality winner is eventually an extreme false positive: top-one selection loses a multiple-comparisons contest that the portfolio, which retains every named box, never enters. The pooled ranking still contains substantial information; converting it into one repeated action is what discards it. The portfolio formula is the asymptotic form of the finite reallocation identity that reappears under copying in section 6: the planner's entire limiting gain over private search, $e^{-\alpha r}\left[1-(1-\alpha)e^{\alpha}\right]$, comes from redirecting duplicated capacity to unnamed states on the event that every clue misses.

EC.6 Computational verification and status of claims

The accompanying script `verify_shared_discovery_v8_6.py` performs the following checks.

1. It verifies the reciprocal-share identity and the water-filling equilibrium on 200 random posteriors, including dominant-posterior pure corners.
2. It verifies the welfare identity $G^{\mathrm{sym}}_{\mathrm{market}} = N\lambda$ posterior by posterior.
3. It reproduces the uniform-posterior formula and the $e/(e-1)$ limit.
4. It verifies the tight $2 - 1/N$ family and the sole-rescue pure-equilibrium characterization.
5. It exactly enumerates the 67 canonical report-count classes for every copying value, verifies that class probabilities sum to one, and reproduces the private-search closed form.
6. It reproduces the independent pooled frontier, the canonical market value 0.599099252439, the full copying table, and the crossover $c^* = 0.788461656521$.

7. It verifies the blind-control values $8/16 = 0.50$ and $1-(1-1/16)^8 = 0.403281$, and that the blind-sentinel hybrid family is maximized at zero sentinels.

8. It computes the average-quality column exactly ($Q^{\mathrm{sym}}_{\mathrm{market}} = 0.344136160273$) and verifies the redundancy identity $NQ - G = \mathbb{E}[(K-1)^+]$ class by class for the market and in closed form for the other protocols.

9. It evaluates the proportional sparse-evidence limits at $\alpha = 1/2$, $r = 3.75$, checks the fill-identity form of the portfolio limit, and runs a seeded Monte Carlo convergence diagnostic at $M \in \{16, 64, 256\}$.

10. It cross-checks a closed-form expression for the common-cue likelihood coefficients against the direct enumeration on three (M, N, p) configurations, and validates a batched equilibrium solver against the scalar solver on random posteriors.

11. It solves the market limit equation equation (EC.6) to high precision ($\Lambda = 1.094021073740$, $G = 0.547010536870$), verifies the tilted-form identity to 10^{-11}, confirms that the active support starts at two reports, and runs a seeded Monte Carlo convergence diagnostic for the finite market at $M \in \{16, 64, 256, 1024\}$.

12. It computes the partial-conformity curve of remark 1 exactly for $a = 0, \ldots, 8$, verifies the endpoints $G(0) = G_{\mathrm{private}}$ and $G(8) = G_{\mathrm{consensus}}$, and confirms the strict monotone decline.

13. It verifies the limit gap decomposition $G_{\mathrm{private}} - G^{\mathrm{sym}}_{\mathrm{market}} \to g_1 + \sum_{j\geq 2} g_j e^{-x_j}$, whose two terms are 0.287541 and 0.012094, to 10^{-11}, and runs a seeded fixed-p Monte Carlo confirming the abundant-evidence remark (consensus $0.385 \to 0.947 \to 1.000$ at $M \in \{16, 64, 256\}$).

14. It scans the 72-combination comparative-statics grid, verifies that the market–private gap changes sign at most once in every combination, and writes the crossover locations to `grid_crossings.csv`.

15. It generates every figure and the underlying CSV tables.

Analytically established. The value-of-information and fixed-budget dominance results; posterior-count ranking under independent symmetric signals; private coverage dominance over one-action consensus; the weak monotone harm of partial conformity by set inclusion; the blind-control values; the redundancy identity $NQ - G = \mathbb{E}[(K-1)^+]$; the proportional sparse-evidence limits, including the market cell of equation (EC.6) and the abundant-evidence consistency remark; the reciprocal-share identity; the exact-potential and payout identities for equal sharing; the anonymous water-filling equilibrium including its pure corner; $G^{\mathrm{sym}}_{\mathrm{market}} = N\lambda$; the exact mixed PoA $= 2 - 1/N$ upper bound and tight family; full pure-strategy implementation under sole-rescue; channel collapse and the closed-form private discovery function; failure of count ranking under

dependence by explicit counterexample; the fill identity; and strict monotonicity of the centralized planner gain.

Exactly enumerated or numerically solved for the canonical model. Consensus, private, planner, and action-budget frontier values; the partial-conformity curve $G(a)$; recovery budget; expected distinct-action counts; the anonymous market value and its average-quality column; the common-cue table; and the market/private crossover. Enumeration over report classes is exact up to floating-point evaluation of finite formulas; one-dimensional equilibrium and crossover equations are solved to high numerical precision.

Computational patterns, not general theorems. The canonical herding discount vanishes for action budgets $L \geq 3$ on the tested copying grid; the canonical recovery budget is nonincreasing over that grid; and the market/private gap changes sign at most once on the tested 72-combination grid—exactly once in 52 combinations and never in the remaining 20, where private report-following stays ahead through $c = 0.98$. These patterns should not be generalized without additional analysis.

References